

D40EN throughput measurement process

1. Copy firmware

Enter the directory where the firmware is stored android_driver/USB/driver_fw/fw/

Copy the folder aic8800D80N to the directory /lib/firmware

sudo cp aic8800D80N /lib/firmware -r

```
aic@aic:~$  
aic@aic:~$ cd android_driver/USB/driver_fw/fw/  
aic@aic:~/android_driver/USB/driver_fw/fw$ sudo cp aic8800D80N/ /lib/firmware/ -r  
aic@aic:~/android_driver/USB/driver_fw/fw$
```

2. Load driver

compile the driver

Enter the android_driver/USB/driver_fw/driver/aic8800 directory and execute the make command

Obtain the aic_load_fw.ko and aic8800_fdrv.ko files

Execute commands in the aic8800 directory

sudo insmod aic_load_fw/aic_load_fw.ko

sudo insmod aic8800_fdrv/aic8800_fdrv.ko

```
aic@aic:~$  
aic@aic:~$ cd android_driver/USB/driver_fw/drivers/aic8800/  
aic@aic:~/android_driver/USB/driver_fw/drivers/aic8800$ make  
make -C /lib/modules/5.15.0-139-generic/build M=/home/aic/android_driver/USB/driver_fw/drivers/aic8800  
0 ARCH=x86_64 CROSS_COMPILE= modules  
make[1]: warning: jobserver unavailable: using -j1. Add '+' to parent make rule.  
make[1]: Entering directory '/usr/src/linux-headers-5.15.0-139-generic'  
make[3]: warning: -j4 forced in makefile: resetting jobserver mode.  
make[1]: Leaving directory '/usr/src/linux-headers-5.15.0-139-generic'  
aic@aic:~/android_driver/USB/driver_fw/drivers/aic8800$ sudo insmod aic_load_fw/aic_load_fw.ko  
aic@aic:~/android_driver/USB/driver_fw/drivers/aic8800$ sudo insmod aic8800_fdrv/aic8800_fdrv.ko  
aic@aic:~/android_driver/USB/driver_fw/drivers/aic8800$
```

ifconfig

Before loading the driver

```
aic@aic:~/android_driver/USB/driver_fw/drivers/aic8800$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.15.10.91 netmask 255.255.255.0 broadcast 10.15.10.255  
    inet6 fe80::240a:f155:7f39:45cb prefixlen 64 scopeid 0x20<link>  
    ether 8c:ec:4b:9d:2a:d2 txqueuelen 1000 (Ethernet)  
    RX packets 121747 bytes 70521367 (70.5 MB)  
    RX errors 0 dropped 4290 overruns 0 frame 0  
    TX packets 35233 bytes 3806855 (3.8 MB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 1260 bytes 128633 (128.6 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 1260 bytes 128633 (128.6 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
wlan0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    ether 28:3a:4d:2b:71:75 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

After loading the driver

```
alcatel@alcatel:~/android_driver/USB/driver_fw/drivers/alcatel$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.15.10.91 netmask 255.255.255.0 broadcast 10.15.10.255
    inet6 fe80::240a:f155:7f39:45cb prefixlen 64 scopeid 0x20<link>
    ether 8c:ec:4b:9d:2a:d2 txqueuelen 1000 (Ethernet)
    RX packets 121747 bytes 70521367 (70.5 MB)
    RX errors 0 dropped 4290 overruns 0 frame 0
    TX packets 35233 bytes 3806855 (3.8 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1260 bytes 128633 (128.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1260 bytes 128633 (128.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

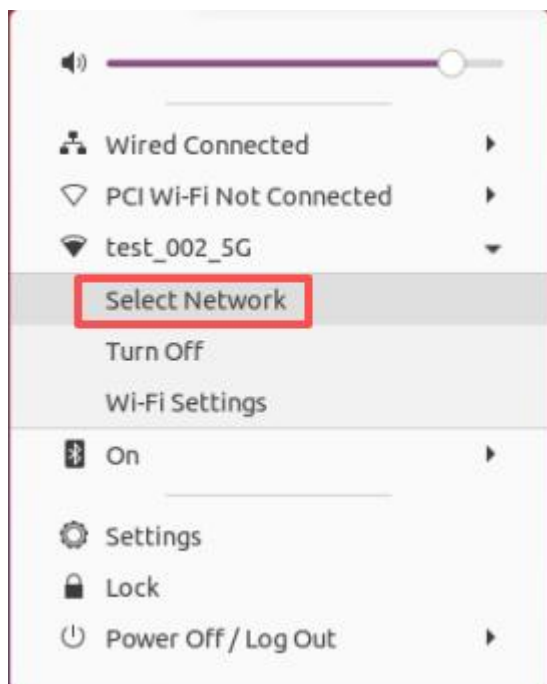
wlan0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 28:3a:4d:2b:71:75 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.75.194 netmask 255.255.255.0 broadcast 192.168.75.255
    inet6 fe80::7a98:1531:12c9:b849 prefixlen 64 scopeid 0x20<link>
    ether 88:00:33:77:74:68 txqueuelen 1000 (Ethernet)
    RX packets 15 bytes 1974 (1.9 KB)
    RX errors 0 dropped 2 overruns 0 frame 0
    TX packets 71 bytes 8532 (8.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

If there is a new WLAN after loading the driver, it indicates that the driver loading was successful

3. Connect to Wi-Fi

Method 1: Connect using Ubuntu graphical interface



Method 2: Connect using the WPA_Supplicant instruction

The first step is to compile WPA_Supplicant

You can download the source code of [hostapd and wpa_supplicant](#)

cd wpa_supplicant-2.10/wpa_supplicant

execute **sudo cp defconfig .config**, and then **make**

```
aic@aic:~/work/wpa_supplicant-2.10/wpa_supplicant$ sudo cp defconfig .config
[sudo] password for aic:
aic@aic:~/work/wpa_supplicant-2.10/wpa_supplicant$ make
CC config.o
CC notify.o
CC bss.o
CC eap_register.o
CC ../src/utls/common.o
CC ../src/utls/config.o
CC ../src/utls/wpa_debug.o
CC ../src/utls/wpabuf.o
CC ../src/utls/bitfield.o
CC ../src/utls/ip_addr.o
CC ../src/utls/crc32.o
CC op_classes.o
CC rrm.o
CC twt.o
CC robust_av.o
CC wmm_ac.o
CC ../src/utls/os_unix.o
CC ../src/utls/eloop.o
CC config_file.o
CC ../src/rsn_supp/wpa_ft.o
CC ../src/common/sae.o
CC ../src/common/dpp.o
CC ../src/common/dpp_auth.o
CC ../src/common/dpp_backup.o
CC ../src/common/dpp_crypto.o
CC ../src/common/dpp_pkex.o
CC ../src/common/dpp_reconfig.o
CC ../src/common/dpp_tcp.o
CC dpp_supplicant.o
CC ../src/rsn_supp/tdls.o
CC ../src/rsn_supp/wpa.o
```

the compilation result is

```
CC ctrl_iface_unix.c
CC dbus/dbus_dict_helpers.c
CC dbus/dbus_new_helpers.c
CC dbus/dbus_new.c
CC dbus/dbus_new_handlers.c
CC dbus/dbus_common.c
CC dbus/dbus_new_handlers_wps.c
CC dbus/dbus_new_handlers_p2p.c
CC dbus/dbus_new_introspect.c
CC ../src/utls/base64.c
CC sme.c
CC ../src/common/ieee802_11_common.c
CC ../src/common/hw_features_common.c
CC ../src/eap_common/eap_common.c
CC ../src/crypto/sha1-prf.c
CC ../src/crypto/sha1-tprf.c
CC ../src/crypto/sha1-tlsprf.c
CC bgscan_simple.c
CC bgscan.c
CC ../src/common/gas_server.c
CC ../src/common/gas.c
CC gas_query.c
CC offchannel.c
CC ../src/utls/json.c
CC ../src/drivers/driver_common.c
CC wpa_supplicant.c
CC events.c
CC bssid_ignore.c
CC wpas_glue.c
CC scan.c
CC main.c
CC ../src/drivers/driver_wired.c
CC ../src/drivers/driver_macsec_linux.c
CC ../src/drivers/driver_wired_common.c
CC ../src/drivers/driver_nl80211.c
CC ../src/drivers/driver_nl80211_capa.c
CC ../src/drivers/driver_nl80211_event.c
CC ../src/drivers/driver_nl80211_monitor.c
CC ../src/drivers/driver_nl80211_scan.c
CC ../src/drivers/netlink.c
CC ../src/drivers/rfkill.c
CC ../src/utls/radiotap.c
CC ../src/drivers/linux_ioctl.c
CC ../src/drivers/driver_wext.c
CC ../src/drivers/drivers.c
CC ../src/l2_packet/l2_packet_linux.c
LD wpa_supplicant
CC wpa_cli.c
CC ../src/common/wpa_ctrl.c
CC ../src/common/cli.c
CC ../src/utls/edit_simple.c
LD wpa_cli
CC wpa_passphrase.c
LD wpa_passphrase
sed systemd/wpa_supplicant.service.in
sed systemd/wpa_supplicant.service.arg.in
sed systemd/wpa_supplicant-nl80211.service.arg.in
sed systemd/wpa_supplicant-wired.service.arg.in
sed dbus/fi.w1.wpa_supplicant1.service.in
a1c@a1c:~/work/wpa_supplicant-2.10/wpa_supplicant$
```

During the compilation process, it may be necessary to install some dependency libraries:

```
sudo apt-get install libnl-3-dev
sudo apt-get install libnl-genl-3-dev
sudo apt install libnl-route-3-dev
sudo apt install libssl-dev
sudo apt install libdbus-1-dev
```

Step 2, use WPA_Supplicant to connect to WiFi

Execute commands in the wpa_supplicant-2.10/wpa_supplicant directory

sudo ./wpa_supplicant -i wlan1 -c ~/conf/wpa_supplicant.conf

```
aic@aic:~/work/wpa_supplicant-2.11/wpa_supplicant$ sudo killall wpa_supplicant
aic@aic:~/work/wpa_supplicant-2.11/wpa_supplicant$ sudo ./wpa_supplicant -i wlan0 -c ~/conf/wpa_supplicant.conf
Successfully initialized wpa_supplicant
nl80211: kernel reports: Registration to specific type not supported
```

Among them, wlan1 is the name of the network card after loading the driver

-Add the directory where the conf file is located after 'c', and the name of the conf file can be customized

wpa_supplicant.conf File Configuration

```
ctrl_interface=/var/run/wpa_supplicant
update_config=1
pmf=1
network={
    ssid="AIC8800-TEST-AP"
    psk="qqqqqqqqq"
}
```

AIC8800-TEST-AP is an SSID that needs to be connected to WiFi

Use the command **wpa_cli -i wlan1 status** to determine if the connection is successful

```
aic@aic:~/work/wpa_supplicant-2.10/wpa_supplicant$ sudo ./wpa_supplicant -i wlan1 -c ~/work/conf/wpa_supplicant.conf -B
Successfully initialized wpa_supplicant
nl80211: kernel reports: Registration to specific type not supported
aic@aic:~/work/wpa_supplicant-2.10/wpa_supplicant$ sudo ./wpa_cli -i wlan1 status
wpa_state=SCANNING
ip_address=192.168.75.194
p2p_device_address=88:00:33:77:74:69
address=88:00:33:77:74:68
uuid=11d1864f-77cc-5341-a521-2533a54927d4
aic@aic:~/work/wpa_supplicant-2.10/wpa_supplicant$
```

wpa_status=SCANNING indicates that SSID is being scanned

```
aic@aic:~/work/wpa_supplicant-2.11/wpa_supplicant$ sudo ./wpa_cli -i wlan0 status
bssid=34:46:ec:63:66:48
freq=5805
ssid=WS7200_5G
id=0
mode=station
wifi_generation=6
pairwise_cipher=CCMP
group_cipher=TKIP
key_mgmt=WPA2-PSK
wpa_state=COMPLETED
p2p_device_address=88:00:33:77:a1:10
address=88:00:33:77:a1:11
uuid=1deabd5b-61b3-5739-a56b-7dbfacd347e0
ieee80211ac=1
```

wpa_status=Completed indicates successful connection

Then use **sudo dhclient -i wlan1-q** to assign an IP address to the network card

```

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1268 bytes 129457 (129.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1268 bytes 129457 (129.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 28:3a:4d:2b:71:75 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    ether 88:00:33:77:74:68 txqueuelen 1000 (Ethernet)
    RX packets 22 bytes 2914 (2.9 KB)
    RX errors 0 dropped 7 overruns 0 frame 0
    TX packets 96 bytes 11081 (11.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

aic@aic:~/work/wpa_supplicant-2.10/wpa_supplicant$ sudo dhclient -i wlan1 -q
aic@aic:~/work/wpa_supplicant-2.10/wpa_supplicant$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.15.10.91 netmask 255.255.255.0 broadcast 10.15.10.255
    inet6 fe80::240a:f155:7f39:45cb prefixlen 64 scopeid 0x20<link>
    ether 8c:ec:4b:9d:2a:d2 txqueuelen 1000 (Ethernet)
    RX packets 123561 bytes 70768703 (70.7 MB)
    RX errors 0 dropped 4397 overruns 0 frame 0
    TX packets 35281 bytes 3810575 (3.8 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1280 bytes 130429 (130.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1280 bytes 130429 (130.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 28:3a:4d:2b:71:75 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.75.194 netmask 255.255.255.0 broadcast 192.168.75.255
    ether 88:00:33:77:74:68 txqueuelen 1000 (Ethernet)
    RX packets 22 bytes 2914 (2.9 KB)
    RX errors 0 dropped 7 overruns 0 frame 0
    TX packets 107 bytes 12731 (12.7 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

aic@aic:~/work/wpa_supplicant-2.10/wpa_supplicant$

```

4. Iperf

Using iperf3 tool to run streams

The PC on the other end is connected to the router through a network cable, and the terminal inputs the command **iperf3- s - i 1**

Test PC terminal input command **iperf3- c xxx.xxx.xx (opposite end IP address) - i 1- t 60**